

EDUCATION

M.E. in Applied Mechanics

Sep 2023 - June 2025 (Expected)

National Taiwan University, Taipei, Taiwan (R.O.C.).

- Focus on Computational Fluid Dynamics (CFD), Tensor Basis Neural Network (TBNN), High-Performance Computing (HPC), Message Passing Interface (MPI).
- Supervisor: Yi-Ju Chou.

B.E. in Civil Engineering

Sep 2019 - Jul 2023

National Taiwan University, Taipei, Taiwan (R.O.C.).

- Focus on Fluid Mechanics, Numerical Simulations, Civil Engineering Practices.

PUBLICATIONS

Tsai, Y.-H., & Chou, Y.-J. (Accepted, Dec 2024).

“On the suspension and deposition within turbidity currents”. *Journal of Fluid Mechanics*.

Tsai, Y.-H., & Cheng, C.-J., & Chou, Y.-J. (In preparation).

“Data-driven turbulence modeling by using Tensor-Based Neural Network with considering geometric effect”.

RESEARCH EXPERIENCE

Research Assistant | *Data-Driven Turbulence Model*

Sep 2023 - Present

National Taiwan University, Taipei, Taiwan (R.O.C.).

- Developed a physics-based Data-Driven Turbulence model (TBNN) based on fluid mechanics equations.
- Integrated TBNN models with CFD solvers (RANS), improving the accuracy of simulations.
- This method improves simulation accuracy, achieving up to a 75% reduction in relative error compared to traditional turbulence models.

Research Assistant | *Turbidity Currents Numerical Simulation*

Sep 2022 - Apr 2024

National Taiwan University, Taipei, Taiwan (R.O.C.).

- Used a Fortran-based CFD framework for particle flow simulation (point particle), modeling the behavior (four-way coupling) of 100 million particles.
- Improved efficiency and reduced runtime with Linked-list structures and MPI parallel computing.
- Investigated turbidity current propagation speed, settling velocity, particle deposition and collision, and overall flow field development.

Cluster Administrator | *High-Performance Computing*

Sep 2022 - Present

National Taiwan University, Taipei, Taiwan (R.O.C.).

- Managed an HPC cluster to support large-scale numerical simulations and AI model training.
- Configured Linux systems for a secure and efficient environment, including virtualization, networking, and resource optimization.
- Resolved hardware and software issues to ensure stability and minimize downtime.

GRANTS & FELLOWSHIPS

Graduate Scholarship

2023(First Place Admission), 2024

National Taiwan University, M.E. in Applied Mechanics, Taipei, Taiwan (R.O.C.).

Research Assistant Fellowship

2023, 2024

National Science and Technology Council (NSTC), Taiwan (R.O.C.).

CONFERENCES ATTENDED

48th Conference on Theoretical and Applied Mechanics (CTAM 2024) Nov 2024

Society of Theoretical and Applied Mechanics of the Republic of China, Hsinchu, Taiwan (R.O.C.).

- **Oral Presentation:** “Data-Driven Turbulence Modeling Using Tensor-Based Neural Network with Consideration of The Geometric Effect”.

45th Ocean Engineering Conference (OEC 2023) Nov 2023

The Taiwan Society of Ocean Engineering, Keelung, Taiwan (R.O.C.).

- **Poster Presentation:** “Numerical Simulation of Turbidity Currents in Lock-Exchange Problem”.

28th National Computational Fluid Dynamics Conference (NCFD 2023) Aug 2023

Aeronautical and Astronautical Society of the Republic of China, Taipei, Taiwan (R.O.C.).

WORK EXPERIENCE

Intern Experience | *Hydraulic Engineering* Jul 2023 - Aug 2023

Stantec Taiwan Inc, Taipei, Taiwan (R.O.C.).

Intern Experience | *On-Site Engineering* Jul 2022 - Aug 2022

New Asia Construction & Development Corp, Taipei, Taiwan (R.O.C.).

ADDITIONAL ACTIVITIES

Competition (4th Place) | *Sustainable Campus Competition, Sponge NTU* Jun 2022 - Dec 2022

National Taiwan University, Taipei, Taiwan (R.O.C.).

Capstone Project | *Small Hydropower Development* Feb 2023 - Jun 2023

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

Capstone Project | *Dam-break Analysis and Countermeasures for Shihmen Reservoir* Feb 2022 - Jun 2022

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

Keystone Project (Teaching Assistant) | *Flying Rotors Design* Sep 2022 - Jan 2023

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

Keystone Project | *Floating Structure Design for Aquatic Buildings* Sep 2021 - Jan 2022

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

Cornerstone Project | *Small-Scale Earthquake-Resistant Tower Structure Design* Feb 2021 - Jun 2021

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

Cornerstone Project | *Marble Track Structure Design* Sep 2020 - Jan 2021

National Taiwan University, B.E. in Civil Engineering, Taipei, Taiwan (R.O.C.).

SKILLS & LANGUAGES

Languages: Mandarin, English.

Computational Fluid Dynamics: ANSYS Fluent, OpenFOAM, FEniCS, In-house Source Code.

Operating Systems: Linux, Windows, macOS.

Programming Design: C++, Python, MATLAB, FORTRAN, Shell Scripts.

Artificial Intelligence: TensorFlow, PyTorch, Keras, Scikit-Learn.

Engineering Drafting: AutoCAD, SketchUp, SolidWorks, Fusion360, Revit.

REFERENCES

Yi-Ju Chou, Professor of Applied Mechanics

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